Do Now · retrieval
Waterfall, Agile and Extreme Programming

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which OCR pseudocode function reads the next line of a text file?

Q2

What is a function parameter?

Q3

Why use boundary-value testing?

Q4

If requirements frequently change, what is the risk of freezing all requirements before implementation?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Waterfall is sequential — Agree requirements, then design, implementation, testing and deployment. Revisiting an earlier stage can be costly.

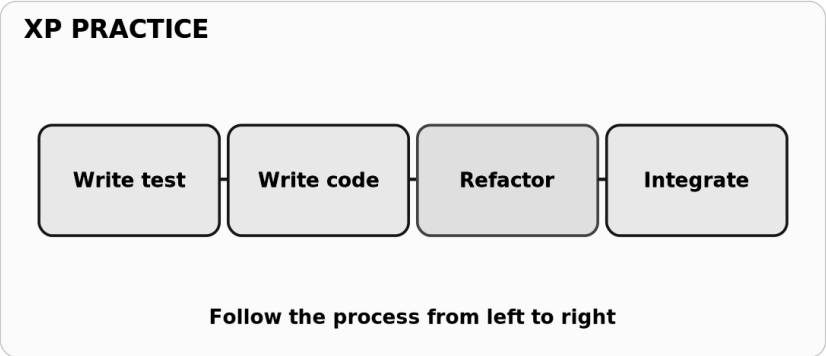
Core idea 2

Agile delivers incrementally — Each short iteration produces working software and seeks feedback, so priorities can change between iterations.

Core idea 3

Extreme Programming (XP) — XP is an agile methodology emphasising practices such as test-driven development, pair programming, refactoring and continuous integration.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Waterfall — Primarily sequential software-development life cycle.

Agile — Iterative approach using feedback and incremental releases.

XP — Extreme Programming, an agile methodology with disciplined engineering practices.

sprint — Short time-boxed development iteration.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

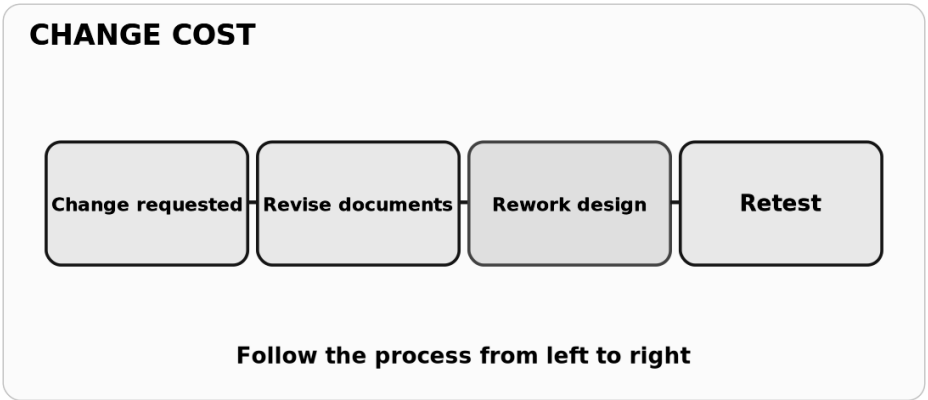
A hospital requests software with a carefully agreed specification and formal approvals. Why might Waterfall suit it?

Follow the worked method

1. Predict first — Requirements and approval milestones can be documented before later stages proceed.

2. Model and explain — Waterfall offers clear documentation and control, but late changes can require expensive rework.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Changing needs and quick feedback favour Agile iterations over a rigid up-front plan.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State a defining characteristic of Waterfall, Agile and XP.

Question 2

A mobile app company receives new customer requirements each fortnight. Recommend a methodology and explain two advantages and one disadvantage.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how test-driven development and continuous integration in XP could reduce defects, and why neither guarantees fault-free software.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Agile means software is never planned or tested.

Use key vocabulary from memory

1. Explain Waterfall in your own words.

2. Explain Agile in your own words.

3. Explain XP in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.